

Nesterov's Accelerated Gradient Descent's Speed

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Introduction

Recent works like **dziugaite2017computing** and **zhou2018non** have shown that one can obtain non-vacuous “computational” PAC-Bayesian bounds on the risk of neural nets. In this work we improve on existing theoretical PAC-Bayesian risk bounds by using data-dependent priors sensitive to the geometrical properties of training and by considering carefully chosen clusters of multiple nets in tandem.

Notation

- $\delta f(x)$ is the set of **subgradients** of the point $x \in \mathbb{R}^n$. Then the subgradient $g \in \delta f(x)$ satisfies $f(x) - f(y) \leq g^\top(x - y)$.
- $e_1, \dots, e_n$ is the canonical basis of $\mathbb{R}^n$.
- $x^* \in \mathbb{R}^n$ is a point such that it is the minima in any optimisation problem, i.e. $f(x^*) = \min_{x \in \mathbb{R}^n} f(x)$. We always assume x^* exists.
- $\|\cdot\|$ is always the Euclidean norm $\|x\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$ for $x \in \mathbb{R}^n$

Fastest theoretical speed for Gradient Descent

Theorem (1)

Let $t \leq \frac{n-1}{2}$ and $\beta > 0$. There exists a β -smooth convex function f such that for any black-box procedure satisfying $x_{t+1} \in \text{Span}(g_1, \dots, g_t)$ with $g_s \in \delta f(x_s)$:

$$\min_{1 \leq s \leq t} f(x_s) - f(x^*) \geq \frac{3\beta \|x_1 - x^*\|^2}{32(t+1)^2}$$

This gives us a bound on the speed of gradient descent of $O(1/t^2)$. We observe that this bound is achieved by Nesterov's Accelerated Gradient Descent

Crafting a β -smooth function f

To construct a suitable function f that satisfies Theorem (1), Bubeck constructs a symmetric and tri-diagonal matrix

$$(A_k)_{i,j} = \begin{cases} 2, & i = j, \ i \leq k \\ -1, & j \in \{i-1, i+1\}, \ i \leq k, \ j \neq k+1 \\ 0, & \text{otherwise} \end{cases}$$

and defines f as:

$$f(x) = \frac{\beta}{8} x^\top A_{2t+1} x - \frac{\beta}{4} x^\top e_1$$

References

Bubeck, Sebastian (2018). *A short proof for Nesterov's momentum*. <https://web.archive.org/web/20210124000724/https://blogs.princeton.edu/imabandit/2018/11/21/a-short-proof-for-nesterovs-momentum/>. Accessed 6 May 2026.

Sutskever, Ilya et al. (17–19 Jun 2013). “On the importance of initialization and momentum in deep learning”. In: *Proceedings of the 30th International Conference on Machine Learning*. Ed. by Sanjoy Dasgupta and David McAllester. Vol. 28. Proceedings of Machine Learning Research 3. Atlanta, Georgia, USA: PMLR, pp. 1139–1147. URL: <https://proceedings.mlr.press/v28/sutskever13.html>.

Descent Lemma

The following lemma establishes the per-step bound for gradient descent on smooth functions. The original paper states the constrained case but for Nesterov's accelerated gradient descent in the smooth function case, we only need the unconstrained version.

Lemma (2)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex and β -smooth function. Let $x, y \in \mathbb{R}^n$ and $x^+ = x - \frac{1}{\beta} \nabla f(x)$. Then the following holds true:

$$f(x^+) - f(y) \leq \nabla f(x)^\top (x - y) - \frac{1}{2\beta} \|\nabla f(x)\|^2$$

Nesterov's accelerated gradient descent

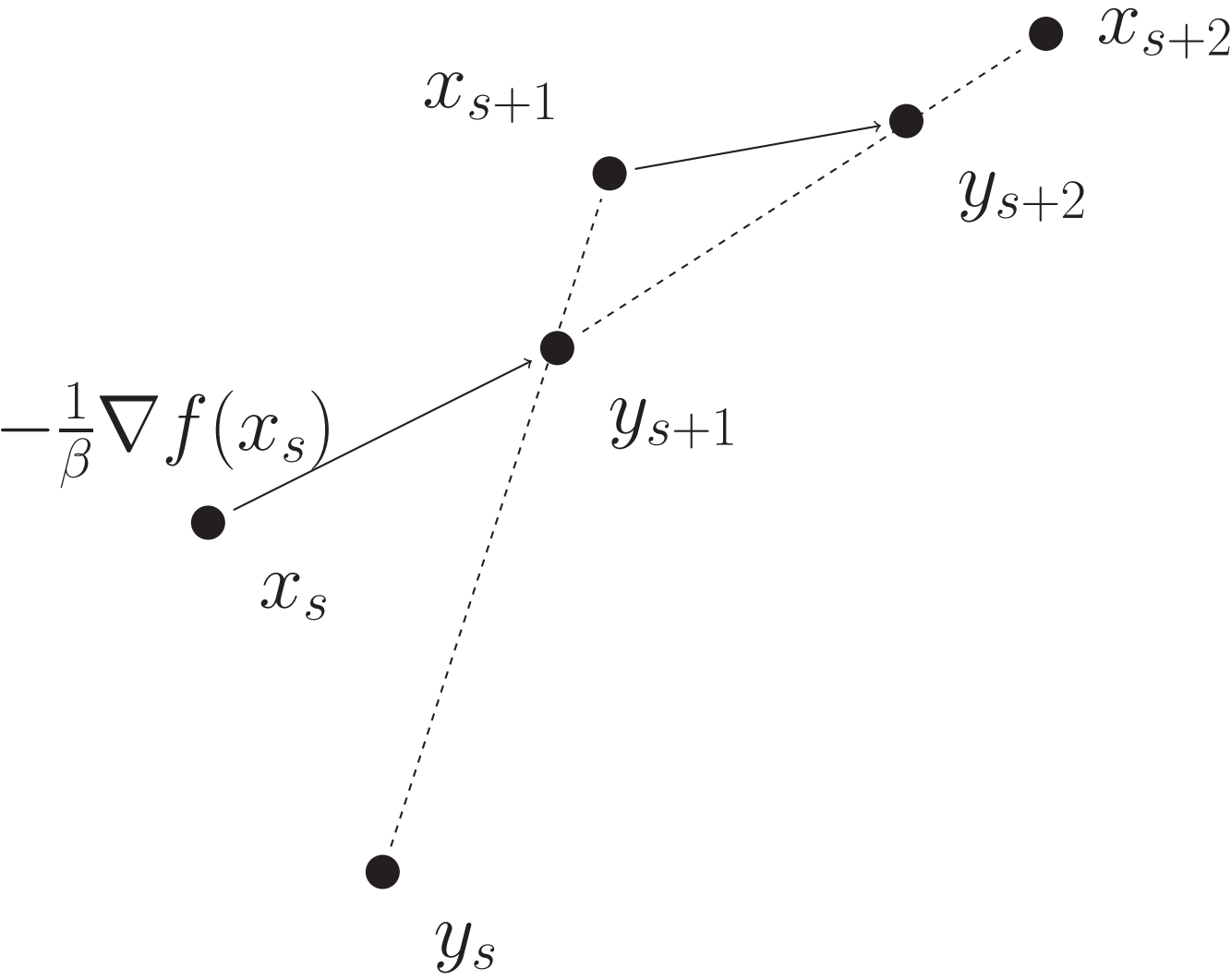


Figure: Illustration of Nesterov's accelerated gradient descent.

First we introduce the following sequences:

$$\lambda_0 = 0, \quad \lambda_t = \frac{1 + \sqrt{1 + 4\lambda_{t-1}^2}}{2}, \quad \text{and} \quad \gamma_t = \frac{1 - \lambda_t}{\lambda_{t+1}}$$

Nesterov's accelerated gradient descent is described as follows. Start at an arbitrary initial point $x_1 = y_1$ and then iterate the following equations for $t \geq 1$

$$y_{t+1} = x_t - \frac{1}{\beta} \nabla f(x_t) \tag{1}$$

$$x_{t+1} = (1 - \gamma_t) y_{t+1} + \gamma_t y_t \tag{2}$$

Nesterov's accelerated gradient descent uses the ideas from classical momentum gradient descent (Sutskever et al. 2013). x_{t+1} can be rewritten as $x_{t+1} = y_{t+1} - \gamma_t(y_{t+1} - y_t)$ which reads as x_{t+1} overshoots y_{t+1} along the vector $(y_{t+1} - y_t)$ with some “momentum” given by γ_t

Theorem (3)

Let f be a convex and β -smooth function, then Nesterov's accelerated gradient descent satisfies

$$f(y_t) - f(x^*) \leq \frac{2\beta \|x_1 - x^*\|^2}{t^2}$$

This gives us the $O(1/t^2)$ complexity. [Write more about it here](#)

Key insights from Proof of Theorem 3

- We can reverse engineer both λ_t and γ_t to show they are not arbitrary. If we rewrite (2) as $x_t = y_t + d_t$ where $d_t = \gamma_t(y_t - y_{t-1})$ is the momentum term and let $g_t = -\frac{1}{\beta} \nabla f(y_t + d_t)$ be the gradient term in y_{t+1} by rewriting x_t as above (1), we get the following two inequalities when applying Lemma 2:

$$f(y_{t+1}) - f(y_t) \leq -\frac{\beta}{2} (\|g_t\|^2 - 2g_t \cdot d_t) \tag{*}$$

$$f(y_{t+1}) - f(x^*) \leq -\frac{\beta}{2} (\|g_t\|^2 - 2g_t \cdot (x_t + d_t - x^*)) \tag{**}$$

- The big idea is to use the identity $\|a\|^2 + 2ab = \|a + b\|^2 - \|b\|^2$ on the combination of both inequalities to obtain a telescopic sum Bubeck 2018. Let $\delta_t = f(y_t) - f(x^*)$. Then by multiplying (*) by $(\lambda_t - 1)$ and adding (**) we obtain:

$$\begin{aligned} \lambda_t \delta_{t+1} - (\lambda_t - 1) \delta_t &\leq -\frac{\beta}{2} (\lambda_t \|g_t\|^2 + 2g_t \cdot (x_t + \lambda_t d_t - x^*)) \\ &= -\frac{\beta}{2} \frac{1}{\lambda_t} (\|x_t + \lambda_t d_t - x^* + \lambda_t g_t\|^2 \\ &\quad - \|x_t + \lambda_t d_t - x^*\|^2) \end{aligned} \tag{†}$$

where the equality comes by applying the identity with $a = \lambda_t g_t$ and $b = x_t + \lambda_t d_t - x^*$.

- To obtain the telescopic sum, we need

$$x_t + \lambda_t d_t - x^* + \lambda_t g_t = x_{t+1} + \lambda_{t+1} d_{t+1} - x^*$$

which comes from observing (†). Since $d_{t+1} = x_{t+1} - y_{t+1} = -\gamma_t(d_t + g_t)$ and $x_{t+1} = x_t + g_t + d_t$:

$$\begin{aligned} x_t + \lambda_t d_t - x^* + \lambda_t g_t &= x_t + g_t + d_t - \lambda_{t+1} \cdot \gamma_t (g_t + d_t) - x^* \\ \lambda_t (d_t + g_t) &= (d_t + g_t) (1 - \lambda_{t+1} \cdot \gamma_t) \end{aligned}$$

and hence $\gamma_t = \frac{1 - \lambda_t}{\lambda_{t+1}}$

- Now to select λ_t , by multiplying λ_t to (†) and denoting $u_t := \frac{\beta}{2} \|x_t + \lambda_t d_t - x^*\|$ we get

$$\lambda_t^2 \delta_{t+1} - (\lambda_t^2 - \lambda_t) \delta_t \leq u_t - u_{t+1}$$

and therefore $\lambda_t^2 - \lambda_t = \lambda_{t-1}^2$ is required to obtain the full telescopic sum. The rest of the proof follows by induction.

Demonstration of stability of the relevant geometrical properties of training

In the *figure to the left* we plot the initial parameter norm $\|B\|_2$ vs the final parameter norm $\|A\|_2$ for training nets of different depths at width 100 on the CIFAR-10 dataset. Thus its demonstrated that the multiplicative factor with which the norm increases during training is fairly stable to architectural changes.

